# Matrix formulations of Bézier technique

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*At the start of this paper, a recurrence formula for calculating the Bézier functions of any order is proved. Based on this formula, the Bézier functions are written in matrix form which enables us to develop the Bézier technique in a concise way. There are some new identities of the Bézier functions which are potentially useful in CAGD. Surprisingly a matrix involving the Bézier functions has been shown to be a doubly-stochastic matrix and the convergence behaviour of its powers is determined. Based upon these facts, the Kelisky–Rivlin theorem for the Bernstein polynomials has been extended to the Bézier curves.*

*Bézier curves, geometry, matrix formulation*

Any parametric curve of degree n was expressed in the form

$$\vec{p}(u) = \sum_{i=0}^{n} f_{n,i}(u)\vec{a}_i \quad (0 \leqslant u \leqslant 1) \tag{1}$$

by Bézier[1] where

$$f_{n,o}(u) \equiv 1$$

$$f_{n,i}(u) = \frac{(-u)^i d^{i-1}}{(i-1)!\, du^{i-1}} \left[\frac{(1-u)^n - 1}{u}\right] \tag{2}$$

are called Bézier functions. $\vec{a}_0, \vec{a}_1, \ldots \vec{a}_n$ are $n + 1$ vectors in the space. The vector $\vec{a}_0$ points to the beginning point of the curve. The polygon formed by $\vec{a}_1, \vec{a}_2, \ldots \vec{a}_n$, which is called Bézier polygon, determines completely the shape of the curve (1). Analytical properties of the Bézier functions have been studied[2,3] and $f_{5,1}$, $f_{5,2}$, $f_{5,3}$, $f_{5,4}$ and $f_{5,5}$ are shown from the top to the bottom respectively in Figure 1.

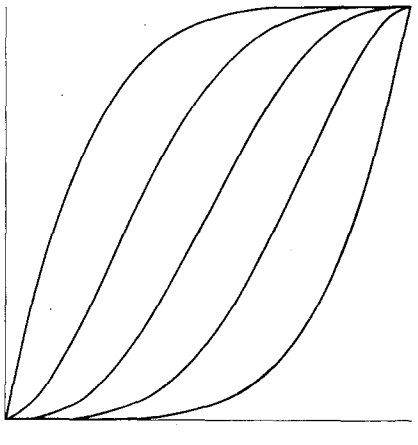

*Figure 1. Bézier functions $f_{5,1} - f_{5,5}$*

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A R Forrest proposed an equivalent way to express the Bézier curves. That is the Bernstein form of the Bézier curves in which instead of the Bézier functions $f_{n,i}$ and the edges of the Bézier polygon, the Bernstein basis functions $J_{n,i}(u)$ and the vertices of the polygon are used. Although the Bernstein form is generally adopted by most workers in computer-aided geometric design, this author thinks that the Bézier form is as convenient to use as the Bernstein form at least in the case of curves. This is due to the very nice mathematical properties of the Bézier functions proved in this paper.

## RECURRENCE FORMULA FOR $f_{n,i}$

We have that

$$f_{n,i}(u) = (1-u)f_{n-1,i}(u) + u\, f_{n-1,i-1}(u) \tag{3}$$

for $i = 1,2, \ldots n$, if we set $f_{n-1,\, n}(u) \equiv 0$

*Proof*

For $i = 1$, by equation (2) we have

$$f_{n,1}(u) = 1 - (1-u)^n = (1-u)\,[1 - (1-u)^{n-1}] + u$$
$$= (1-u)f_{n-1,1}(u) + u\, f_{n-1,0}(u)$$

For $2 \leqslant i \leqslant n$, since

$$\frac{(1-u)^n - 1}{u} = \frac{(1-u)[(1-u)^{n-1} - 1]}{u} - 1$$

then by the Leibnitz' formula for higher-order derivatives of the product of two functions, we have

$$f_{n,i}(u) = \frac{(-u)^i}{(i-1)!}\frac{d^{i-1}}{du^{i-1}}\,(1-u)\cdot\left[\frac{(1-u)^{n-1} - 1}{u}\right]$$

$$= \frac{(-u)^i}{(i-1)!}$$

$$\left[(1-u)\frac{d^{i-1}}{du^{i-1}}\frac{(1-u)^{n-1} - 1}{u} - \binom{i-1}{1}\frac{d^{i-2}}{du^{i-2}}\frac{(1-u)^{n-1} - 1}{u}\right]$$

$$= (1-u)f_{n-1,i}(u) + u\, f_{n-1,i-1}(u)$$

This completes the proof of equation (3).

This means that based on $f_{1,0}(u) = 1$ and $f_{1,1}(u) = u$, all of the other $f_{n,i}(u)$ can be uniquely determined by equation (3). It is pointed out that (3) can be rewritten in the form as

$$u[f_{n-1,i-1}(u) - f_{n-1,i}(u)] = f_{n,i}(u) - f_{n-1,i}(u) \tag{4}$$

where $i = 1,2, \ldots n$

### Matrix representation for $f_{n,i}$

Equation (3) enables us to write:

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \\ f_{11} \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \\ u \end{bmatrix} = \begin{bmatrix} u & 1-u & & & \\ & u & 1-u & & \\ & & \ddots & \ddots & \\ & & & & 1-u \\ & & & & u \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \\ f_{21} \\ f_{22} \end{bmatrix} = \begin{bmatrix} u & 1-u & & \\ & u & 1-u & \\ & & & 1-u \\ & & & u \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ f_{11} \end{bmatrix}$$

If the following notations

$$F_n = F_n(u) = \begin{bmatrix} f_{n,1}(u) \\ \vdots \\ f_{n,n}(u) \end{bmatrix}$$

$I = I_n$ the $n \times n$ identity matrix

$$E = \begin{bmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & & 1 \\ & & & & 0 \end{bmatrix}$$

$B = B(u) = uI + (1-u)E$

and $1_n$ the $n \times 1$ matrix whose entries are all 1 have been used, then we have

$$F_n = B^n 1_n \tag{5}$$

which is called the matrix representation of the Bézier functions. Equation (5) shows that $f_{n,i}(u)$ is equal to the sum of all entries in the $i$th row of the matrix $B^n(u)$. In the general case we have that

$$\begin{bmatrix} 1_{n-k} \\ F_k \end{bmatrix} = B^k 1_n \quad k = 0,1,\ldots n \tag{6}$$

and then by equations (5) and (6)

$$F_n = B^{n-k} \begin{bmatrix} 1_{n-k} \\ F_k \end{bmatrix} \tag{7}$$

which establishes the relations between the Bézier functions of different orders.

Direct calculations indicate that

$$E^2 = \begin{bmatrix} 0 & 0 & 1 & & & \\ & 0 & 0 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 0 & 0 & 1 \\ & & & & 0 & 0 \\ & & & & & 0 \end{bmatrix}$$

$$E^{n-1} = \begin{bmatrix} 0 & \ldots & 0 & 1 \\ & \ddots & & 0 \\ & & \ddots & \\ & & & 0 \end{bmatrix} \tag{8}$$

$$E^n = 0$$

The last equality means that E is a nilpotent matrix. We have by this that

$$F_n(0) = [B(0)]^n 1_n = E^n 1_n = 0 1_n = 0$$

$$F_n(1) = [B(1)]^n 1_n = I^n 1_n = 1_n$$

Thus

$$f_{n,i}(0) = 0, \; f_{n,i}(1) = 1 \tag{9}$$

where $i = 1,2,\ldots n$

We have by binomial expansion and $E^n = 0$ that

$$B^n = [uI + (1-u)E]^n$$

$$= \sum_{i=1}^{n} J_{n,i}(u) E^{n-i} \tag{10}$$

where

$$J_{n,i}(u) = \binom{n}{i} u^i (1-u)^{n-i} \tag{11}$$

where $i = 0,1,2,\ldots n$

These are called the Bernstein basis functions. $J_{5,i}(u)$, $i = 0,1\ldots 5$, are shown in Figure 2. It is possible to rewrite equation (10) in more explicit form

$$B^n = \begin{bmatrix} J_{n,n} & J_{n,n-1} & \cdots & J_{n,1} \\ & J_{n,n} & J_{n,n-1} & \vdots \\ & & \ddots & J_{n,n-1} \\ & & & J_{n,n} \end{bmatrix} \tag{12}$$

From equations (5) and (12) we obtain

$$f_{n,i}(u) = J_{n,i}(u) + J_{n,i+1}(u) + \ldots + J_{n,n}(u) \tag{13}$$

for $i = 1,2,\ldots n$

Or

$$f_{n,i}(u) - f_{n,i+1}(u) = J_{n,i}(u) \tag{14}$$

for $i = 0,1,2,\ldots n$

Hence by equation (4) we have

$$f_{n,i}(u) - f_{n-1,i}(u) = u J_{n-1,i-1}(u) \tag{15}$$

for $i = 1,2,\ldots n$

## DERIVATIVES

Taking the first order derivative of $B(u)$ we get

$$B'(u) = \frac{dB}{du} = I - E$$

This is a $n \times n$ matrix with constant entries and is denoted by G. Since $G = I - E$, then it is commutative with $B = uI + (1-u)E$, hence we have

$$\frac{d}{du} B^n = n \frac{dB}{du} B^{n-1} = nG\,B^{n-1}$$

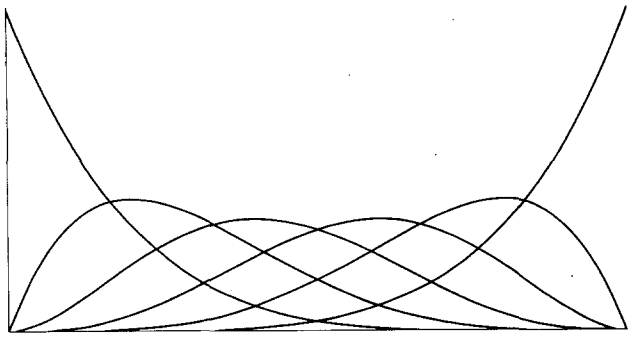

*Figure 2. Bernstein base functions $J_{5,0}(u) - J_{5,5}(u)$*

and then

$$\frac{d}{du} F_n = n\, G \begin{bmatrix} 1 \\ \\ F_{n-1} \end{bmatrix} \tag{16}$$

For higher order derivatives we have

$$\frac{d^\gamma}{du^\gamma} F_n = \frac{n!}{(n-\gamma)!} G^\gamma \begin{bmatrix} 1_\gamma \\ \\ F_{n-\gamma} \end{bmatrix} \tag{17}$$

where $\gamma = 0,1, \ldots n$

We rewrite equation (16) by equation (14) into

$$f'_{n,i}(u) = nJ_{n-1,i-1}(u) \tag{18}$$

where $i = 1,2, \ldots n$

Since $J_{n-1,i-1}(u) > 0$ for $u \in (0,1)$ then we can summarize equations (18), (14) and (15) in the following theorem.

*Theorem*

For fixed $n$ and $i$, $f_{n,i}(u)$ is increasing strictly from 0 to 1 on $[0,1]$.

For fixed $n$ and $u \in (0,1)$, $f_{n,i}(u)$ is decreasing strictly for $i = 1,2, \ldots n$

For fixed $i$ and $u \in (0,1)$, $f_{n,i}(u)$ is increasing strictly for $n = i, i+1, i+2, \ldots$

The first two conclusions have been stated by Chang and Wu[3], but the third is new.

## BÉZIER CURVE

Putting $[\vec{a}] = [\vec{a}_1 \ldots \vec{a}_n]$, then the Bézier curve, equation (1), can be written as

$$\vec{p}(u) = \vec{a}_0 + [\vec{a}]\, F_n \tag{19}$$

It is easy to show the inverse of G

$$G^{-1} = (I - E)^{-1} = I + E + \ldots + E^{n-1} = \begin{bmatrix} 1 & 1 \ldots 1 \\ & 1 \ldots 1 \\ & \ddots \quad \vdots \\ & \quad 1 \end{bmatrix}$$

then we have by (14) that

$$[\vec{a}]\, F_n =$$

$$[\vec{a}]\, G^{-1} G\, F_n = [\vec{a}_1, \vec{a}_1 + \vec{a}_2, \ldots \vec{a}_1 + \vec{a}_2 + \ldots + \vec{a}_n] \begin{bmatrix} J_{n,1} \\ J_{n,2} \\ \vdots \\ J_{n,n} \end{bmatrix}$$

$$= \sum_{i=1}^{n} \left[ \sum_{j=1}^{i} \vec{a}_j \right] J_{n,i}(u)$$

Since $\vec{a}_0 = \sum_{i=0}^{n} J_{n,i}(u)\, \vec{a}_0$ then we have

$$\vec{p}(u) = \sum_{i=0}^{n} \left[ \sum_{j=0}^{i} \vec{a}_j \right] J_{n,i}(u) \tag{20}$$

i.e. the Bernstein form of curve, equation (1).

Consider the $\gamma$th derivative of the Bézier curve. By equation (17) we obtain that

$$\vec{p}^{(\gamma)}(u) = \frac{n!}{(n-\gamma)!} [\vec{a}]\, G^\gamma \begin{bmatrix} 1_\gamma \\ \\ F_{n-\gamma} \end{bmatrix} \tag{21}$$

That is

$$\vec{p}^{(\gamma)}(u) = \frac{n!}{(n-\gamma)!} \left[ \Delta^{\gamma-1} \vec{a}_1 + \sum_{k=1}^{n-\gamma} f_{n-\gamma,k}(u)\, \Delta^\gamma \vec{a}_k \right] \tag{22}$$

for $\gamma = 1,2, \ldots n$, where $\Delta$ represents the difference operator: $\Delta\vec{a}_k = \vec{a}_{k+1} - \vec{a}_k$.

Particularly, we obtain from equation (22) that

$$\vec{p}^{(\gamma)}(0) = \frac{n!}{(n-\gamma)!} \Delta^{\gamma-1} \vec{a}_1 \tag{23}$$

$$\vec{p}^{(\gamma)}(1) = \frac{n!}{(n-\gamma)!} \Delta^{\gamma-1} \vec{a}_{n-\gamma+1} \tag{24}$$

these two formulas can be found in Forrest[2].

## DEGREE ELEVATION

Degree elevation is accomplished by matrix inversion in Bézier[1] (pp 135–136). Here is an explicit formula for degree elevation. Since

$$f_{n+1,i}(u) - f_{n,i}(u) = uJ_{n,i-1}(u) = \frac{i}{n+1} J_{n+1,i}(u)$$

$$= \frac{i}{n+1} [f_{n+1,i}(u) - f_{n+1,i+1}(u)]$$

then

$$f_{n,i}(u) = \left(1 - \frac{i}{n+1}\right) f_{n+1,i}(u) + \frac{i}{n+1} f_{n+1,i+1}(u)$$

for $i = 1,2, \ldots n$

The last $n$ equalities can be written into matrix form

$$F_n = \frac{1}{n+1} T_{n,n+1} F_{n+1}$$

Where T is an $n \times (n+1)$ matrix

$$T_{n,n+1} = \begin{bmatrix} n & 1 & & & \\ & n-1 & 2 & & \\ & & n-2 & 3 & \\ & & & \ddots & \ddots \\ & & & 1 & n \end{bmatrix}$$

Hence if we set

$$\vec{b}_0 = \vec{a}_0$$

$$[\vec{b}] = [\vec{b}_1 \ldots \vec{b}_n \vec{b}_{n+1}] = \frac{1}{n+1} [\vec{a}]\, T_{n,n+1}$$

then we see that $(n+1)$-gon $[\vec{b}]$ and $n$-gon $[\vec{a}]$ determine the same Bézier curve. As an example, consider $n = 3$. In this case

$$T_{3,4} = \begin{vmatrix} 3 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 3 \end{vmatrix}$$

and $[\vec{b}] = \frac{1}{4}[\vec{a}]T_{3,5}$ gives

$$\vec{b}_1 = \frac{3}{4}\vec{a}_1$$

$$\vec{b}_2 = \frac{1}{4}\vec{a}_1 + \frac{1}{2}\vec{a}_2$$

$$\vec{b}_3 = \frac{1}{2}\vec{a}_2 + \frac{1}{4}\vec{a}_3$$

$$\vec{b}_4 = \frac{3}{4}\vec{a}_3$$

4-gon $\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4$ and 3-gon $\vec{a}_1, \vec{a}_2, \vec{a}_3$ determine the same Bézier curve as shown in Figure 3.

In general, the $(n+p)$-gon

$$\frac{n!}{(n+p)!}[\vec{a}]T_{n,n+1}T_{n+1,n+2}\ldots T_{n+p-1,n+p}$$

determines the same curve (1) for $p = 1,2,3,\ldots$

## REPROVING THE THEOREM FOR PLOTTING BEZIER CURVE

In his book, Bézier[1] suggested an interesting method to determine the $\vec{p}(u)$ corresponding to the parameter $u$ on the curve (1).

Starting from the Bézier polygon, $[\vec{a}^0] \equiv [\vec{a}] = [\vec{a}_1 \ldots \vec{a}_n]$, to find a point dividing each directed side in the ratio $u:(1-u)$ and then to join the consecutive points to form a $(n-1)$-gon, $[\vec{a}^1] = [\vec{a}_1^1\ \vec{a}_2^1 \ldots \vec{a}_{n-1}^1]$. Repeating the similar procedure to the $(n-1)$-gon $[\vec{a}^1]$ to obtain the $(n-2)$-gon $[\vec{a}^2] = [\vec{a}_1^2\ \vec{a}_2^2 \ldots \vec{a}_{n-2}^2]$ ... After $n-1$ steps, a directed line segment $\vec{a}_1^{(n-1)}$ is obtained. Bézier asserts that:

- the point dividing $\vec{a}_1^{(n-1)}$ in the ratio $u:(1-u)$ is the point $\vec{p}(u)$ on the curve (1)
- the vector $\vec{a}_1^{(n-1)}$ is the tangent at this point

Figure 4 illustrates the Bézier construction of the point $\vec{p}(\frac{1}{4})$ on a Bézier curve of fourth degree. A proof of the plotting theorem is given by matrix computation[3]. A simpler proof follows.

It is clear by the procedure mentioned above that

$$[\vec{a}]B^k = [* \ldots * \vec{a}^k] \tag{25}$$

for $k = 0,1,\ldots n-1$ and where $*\ldots*$ stands for $k$ entries which we need not be concerned about, and that the first statement of the theorem is equivalent to

$$\vec{p}(u) = \vec{a}_0 + u\sum_{i=0}^{n-1}\vec{a}_1^i \tag{26}$$

Since

$$[\vec{a}]F_n = [\vec{a}]B\,B^{n-1}1_n$$

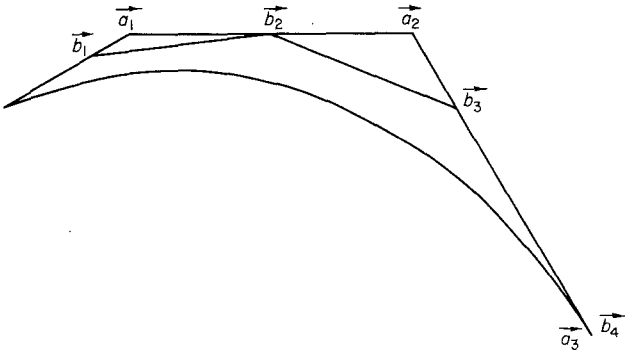


*Figure 3. Bézier curve*

$$= [u\vec{a}_1, \vec{a}^1]\begin{bmatrix} 1 \\ \\ F_{n-1} \end{bmatrix}$$

$$= u\vec{a}_1 + [\vec{a}^1]F_{n-1}$$

$$= u\vec{a}_1 + u\vec{a}_1^1 + [\vec{a}^2]F_{n-2}$$

$$= u\vec{a}_1 + u\vec{a}_1^1 + \ldots + u\,\vec{a}_1^{(n-2)} + [\vec{a}^{(n-1)}]F_1$$

$$= u\sum_{i=0}^{n-1}\vec{a}_1^i$$

then by $\vec{p}(u) = \vec{a}_0 + [\vec{a}]F_n$, equation (26) follows immediately.

In order to prove the second statement, we first go further. Thus we prove a formula other than in equation (22) for the derivative $p^{(\gamma)}(u)$. Since

$$\vec{p}^{(\gamma)}(u) = \frac{d^\gamma}{du^\gamma}[\vec{a}]B^n 1_n = \frac{n!}{(n-\gamma)!}[\vec{a}]G^\gamma B^{n-\gamma}1_n$$

$$= \frac{n!}{(n-\gamma)!}[\vec{a}]B^{n-\gamma}G^\gamma 1_n$$

(where the commutativity of G and B is used) and

$$[\vec{a}]B^{n-\gamma} = [* \ldots \vec{a}_1^{n-\gamma} \ldots \vec{a}_\gamma^{n-\gamma}]$$

$$G^\gamma 1_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ (-1)^{\gamma-1}\begin{bmatrix}\gamma-1\\ \gamma-1\end{bmatrix} \\ \vdots \\ \begin{bmatrix}\gamma-1\\ 1\end{bmatrix} \\ \begin{bmatrix}\gamma-1\\ 0\end{bmatrix} \end{bmatrix} \begin{matrix} \left.\vphantom{\begin{matrix}0\\ \vdots\\ 0\end{matrix}}\right\} n-\gamma \text{ zero} \\ \\ \\ \\ \\ \end{matrix}$$

then

$$\vec{p}^{(\gamma)}(u) = \frac{n!}{(n-\gamma)!}\sum_{i=0}^{\gamma-1}(-1)^{\gamma-1-i}\begin{bmatrix}\gamma-1\\ i\end{bmatrix}\vec{a}_{i+1}^{n-\gamma}$$

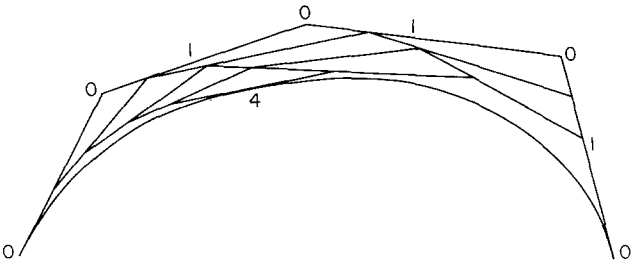


*Figure 4. Bézier construction of $\vec{p}(1/4)$ on a Bézier curve of fourth degree*

for $\gamma = 1,2,\ldots,n$

$$\vec{p}^{(\gamma)}(u) = \frac{n!}{(n-\gamma)!}\,\Delta^{\gamma-1}\,\vec{a}_1^{\,n-\gamma} \tag{27}$$

by the formula (2.7.5) in Davis[4] (p 51). Taking $\gamma = 1$ into equation (27) we have $\vec{p}'(u) = n\,\vec{a}_1^{\,(n-1)}$, hence the second statement of the theorem is proved.

From the proof we see that $\vec{a}_1^{\,k}$ is the hodograph of the Bézier curve determined by $(k+1)$-gon $[\vec{a}_1\ \vec{a}_2\ \ldots\ \vec{a}_{k+1}]$, $k = 0,1,\ldots n-1$.

Equation (26) can be regarded as a representation of the Bézier curve. The representation has a permanence property in the sense of adding a new side to the original polygon to form a new Bézier curve can be accomplished by adding one more term to equation (26).

## A DOUBLY-STOCHASTIC MATRIX INVOLVING THE BÉZIER FUNCTIONS

From equations (9) and (18) we obtain

$$\sum_{i=1}^{n} f_{n,i}(u) = n \sum_{i=1}^{n} \int_0^u J_{n-1,i-1}(t)\,dt$$

$$= n \int_0^u \sum_{i=1}^{n} J_{n-1,i-1}(t)\,dt = n \int_0^u 1\,dt$$

Thus

$$\sum_{i=1}^{n} f_{n,i}(u) = n\,u \tag{28}$$

which is useful in the sequel.

We define an $n \times n$ matrix D with its $(i,j)$ entry $[f_{n,i}(\frac{j}{n}) - f_{n,i}(\frac{j-1}{n})]$. It is interesting to note that **D** is a doubly-stochastic matrix, ie a square matrix with non-negative entries and all its row sums and all its column sums are equal to 1.

Since $f_{n,i}(u)$ is strictly increasing on $[0,1]$, then we have

$$f_{n,i}(\tfrac{j}{n}) - f_{n,i}(\tfrac{j-1}{n}) > 0 \text{ for } 1 \leqslant i, j \leqslant n$$

We have also that all its row sums

$$\sum_{j=1}^{n} [f_{n,i}(\tfrac{j}{n}) - f_{n,i}(\tfrac{j-1}{n})] = f_{n,i}(1) - f_{n,i}(0) = 1,\ 1 \leqslant i \leqslant n$$

and that all its column sums

$$\sum_{i=1}^{n} [f_{n,i}(\tfrac{j}{n}) - f_{n,i}(\tfrac{j-1}{n})] = \sum_{i=1}^{n} f_{n,i}(\tfrac{j}{n}) - \sum_{i=1}^{n} f_{n,i}(\tfrac{j-1}{n})$$

$$= n(\tfrac{j}{n}) - n(\tfrac{j-1}{n}) = 1$$

by equation (28). So it is proved that **D** is an $n \times n$ doubly-stochastic matrix.

The following determines the behaviour of $\mathbf{D}^k$ when $k \to \infty$.

*Theorem*

For fixed $n$, we have that

$$\lim_{k \to \infty} \mathbf{D}^k = \mathbf{K} \tag{29}$$

where **K** is the $n \times n$ matrix with all entries equal to $1/n$.

*Proof*

From the definition of **D** and equation (13) we see that

$$\mathbf{D} = [f_{n,i}(\tfrac{j}{n})]\mathbf{G}$$

$$= \mathbf{G}^{-1}\,[J_{n,i}(\tfrac{j}{n})]\mathbf{G}$$

$$= \mathbf{G}^{-1}\mathbf{T}\,\mathbf{G}$$

The $n \times n$ matrix, **T**, can be partitioned as

$$\mathbf{T} = \begin{bmatrix} \mathbf{Q} & 0 \\ \mathbf{a} & 1 \end{bmatrix}$$

where **Q** is the $(n-1) \times (n-1)$ matrix

$$\mathbf{Q} = \begin{bmatrix} J_{n,1}(\frac{1}{n}), \ldots\ldots J_{n,1}(\frac{n-1}{n}) \\ \vdots \\ J_{n,n-1}(\frac{1}{n}), \ldots J_{n,n-1}(\frac{n-1}{n}) \end{bmatrix}$$

It is clear that each column sum of **Q** is less than 1, so that the absolute value of each eigenvalue of **Q** is less than 1. Hence there exists a matrix **U** such that

$$\mathbf{T} = \mathbf{U}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & \mathbf{C} \end{bmatrix} \mathbf{U}$$

where **C** is an $(n-1) \times (n-1)$ matrix, in which each eigenvalue is less than 1 by the absolute value. Setting $\mathbf{R} = \mathbf{UG}$ then we have

$$\mathbf{D}^k = \mathbf{R}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & \mathbf{C}^k \end{bmatrix} \mathbf{R} \tag{30}$$

Letting $k \to \infty$ in both sides of equation (30), since $\mathbf{C}^k \to 0$, then we have

$$\lim_{k \to \infty} \mathbf{D}^k = \mathbf{R}^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{R}$$

$$= \begin{bmatrix} \lambda_1\mu_1 & \dots & \lambda_1\mu_n \\ \dots & & \\ \lambda_n\mu_1 & \dots & \lambda_n\mu_n \end{bmatrix} \tag{31}$$

where $[\lambda_1 \dots \lambda_n]'$ is the first column of $\mathbf{R}^{-1}$ and $[\mu_1 \dots \mu_n]$ is the first row of $\mathbf{R}$. Since $\mathbf{D}^k$ $(k = 1,2,\dots)$ and $\lim_{k\to\infty} \mathbf{D}^k$ are still doubly-stochastic matrices, then we have from equation (31) that

$$\lambda_1 = \lambda_2 = \dots = \lambda_n$$

$$\mu_1 = \mu_2 = \dots = \mu_n$$

and $\lambda_i\mu_j = 1/n$ for $i,j = 1,2,\dots n$

The theorem is proved.

## KELISKY–RIVLIN THEOREM FOR THE BÉZIER CURVE

In 1967, Kelisky and Rivlin[5] studied the iteration of the Bernstein operators. One of their interesting results is the following.

Let $\mathbf{B}_n g(x)$ be the Bernstein polynomial of degree $n$ of the function $g(x)$, for fixed $n$ we have that $\lim_{k\to\infty} \mathbf{B}_n^k g(x) = g(0) + [g(1) - g(0)]x$ uniformly on $[0,1]$. The theorem means geometrically that the iterates $\mathbf{B}_n^k g(x)$ converge to the line segment joining $[0,g(0)]$ to $[1,g(1)]$ as $k \to \infty$. Their proof of the theorem involved the evaluations of the eigenvalues and eigenvectors of certain matrices. In 1976, Nielson, Riesenfeld and Weiss[6] reproved the theorem by adopting some ideas and theorems from Markov process.

We now prove the Kelisky and Rivlin theorem for the Bézier curve (1). Without loss of generality, it is assumed that $\vec{a}_0 = 0$. We compute $n + 1$ points on the curve: $\vec{p}(\frac{j}{n}), j = 0,1,\dots n$, and form the Bézier polygon with $n$ sides

$$\vec{p}(\frac{j}{n}) - \vec{p}(\frac{j-1}{n}) = \sum_{i=1}^{n} \vec{a}_i \left[f_{n,i}(\frac{j}{n}) - f_{n,i}(\frac{j-1}{n})\right]$$

for $j = 1,2,\dots n$. This polygon determines a new Bézier curve denoted by

$$\mathbf{B}_n[\vec{p}(u)] = \sum_{j=1}^{n} \sum_{i=1}^{n} \vec{a}_i \left[f_{n,i}(\frac{j}{n}) - f_{n,i}(\frac{j-1}{n})\right] f_{n,j}(u)$$

We rewrite the above expression into matrix form

$$\mathbf{B}_n[\vec{p}(u)] = [\vec{a}]\,\mathbf{D}\mathbf{F}_n \tag{32}$$

Comparing equation (32) with equation (1), it can be said that after such a procedure has been done, the matrix $\mathbf{D}$ is inserted between the row matrix formed by the $n$ sides of the Bézier polygon and the column matrix formed by $f_{n,i}$. Then the same procedure is repeated $k$ times, the Bézier curve obtained will be

$$\mathbf{B}_n^k[\vec{p}(u)] = [\vec{a}]\,\mathbf{D}^k\mathbf{F}_n$$

The problem of the convergence behaviour of $\mathbf{B}_n^k[\vec{p}(u)]$ has been shifted to examining the convergence behaviour of $\mathbf{D}^k$ as $k \to \infty$. By the theorem in the previous section and equation (28) we obtain

$$\lim_{k\to\infty} \mathbf{B}_n^k[\vec{p}(u)] = [\vec{a}_1 \dots \vec{a}_n] \begin{bmatrix} \sum_{i=1}^{n} \frac{1}{n} f_{n,i}(u) \\ \\ \sum_{i=1}^{n} \frac{1}{n} f_{n,i}(u) \end{bmatrix}$$

$$= [\vec{a}_1 \dots \vec{a}_n] \begin{bmatrix} u \\ \vdots \\ u \end{bmatrix}$$

$$= u(\vec{a}_1 + \dots + \vec{a}_n)$$

When $u$ varies from 0 to 1, $u(\vec{a}_1 + \dots + \vec{a}_n)$ represents a line segment joining $\vec{p}(0)$ to $\vec{p}(1)$. The Kelisky and Rivlin theorem for the Bézier curve has been proved.


## ACKNOWLEDGEMENT

The work was done when the author was a Visiting Scholar at the Department of Mathematics, University of Utah, USA. The hospitality and cooperation of his host, Professor R E Barnhill, are greatly appreciated. The author thanks Dr Z Shan very much for his helpful suggestions.


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